

CSE/ISyE 6740
Computational Data Analysis

Support Vector Machine

09/21/2026

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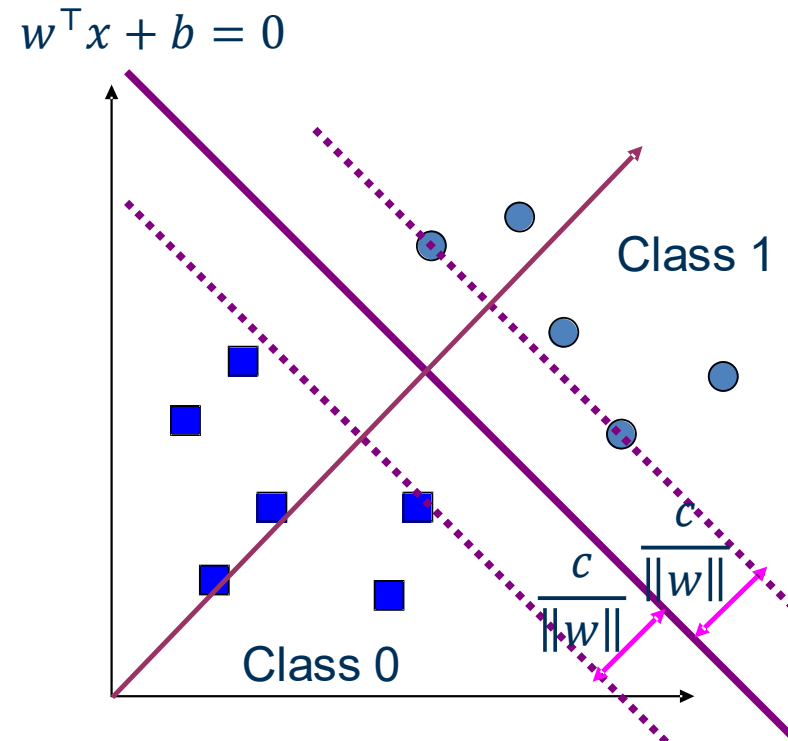
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Soft Margin SVM

Hard-margin SVM

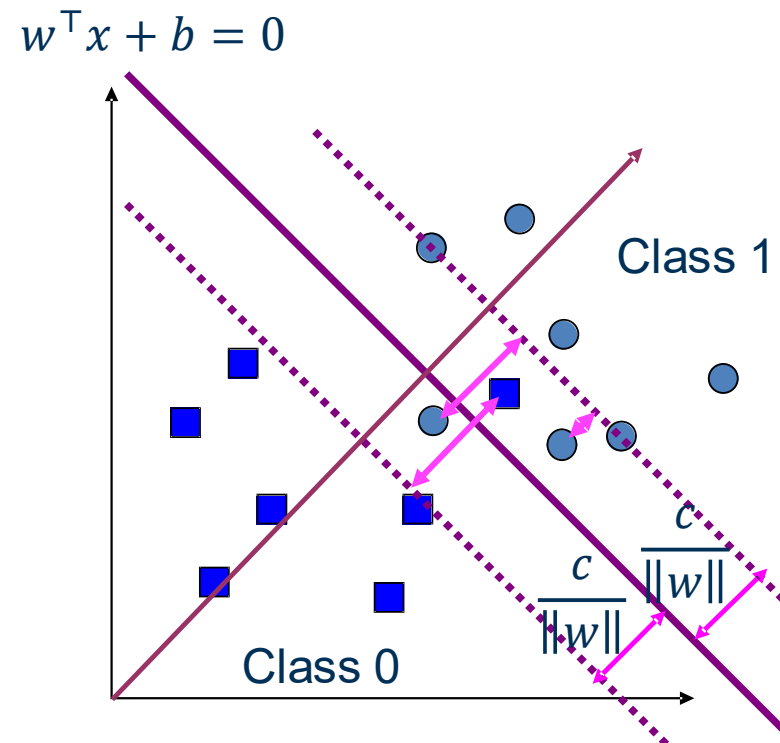
- Find decision boundary w as far from data point as possible

$$\min_{w,b} \frac{1}{2} \|w\|^2$$
$$s.t. y^i(w^\top x^i + b) \geq 1, \quad \forall i$$



Soft-margin SVM

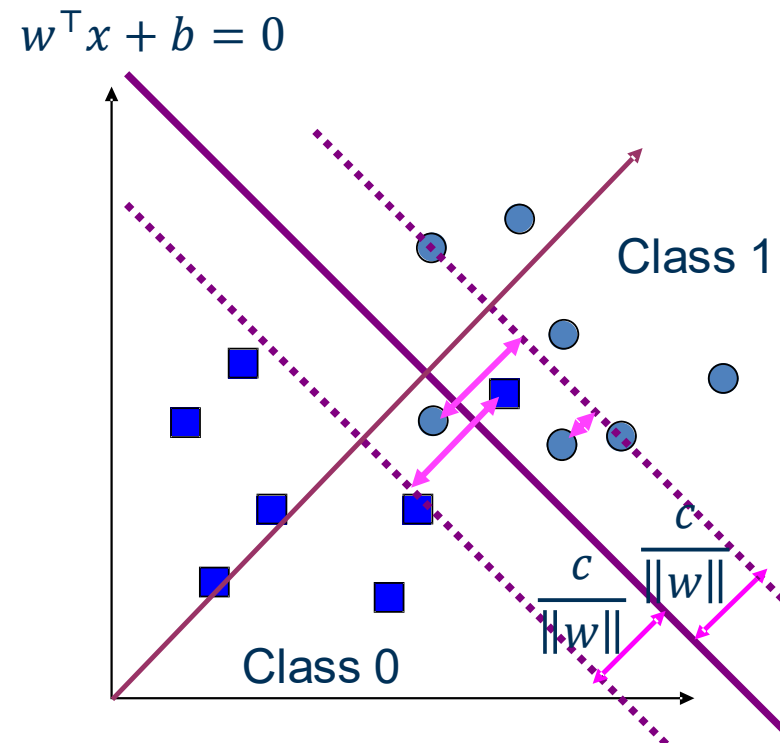
- What if the data is **NOT** linearly separable?
- We will allow points to violate the hard-margin constraint



Soft-margin SVM

- Find decision boundary w as far from data point as possible (with some exceptions ξ)

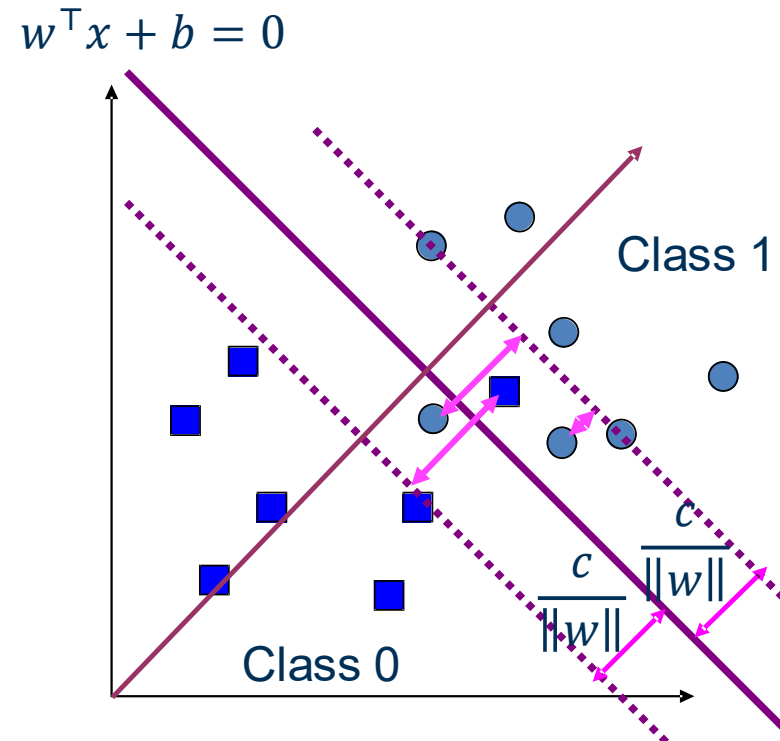
$$\min_{w,b,\xi} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi^i$$
$$s.t. y^i(w^\top x^i + b) \geq 1 - \xi^i, \quad \xi^i \geq 0 \quad \forall i$$



Soft-margin SVM

- Express the optimization problem in standard form:

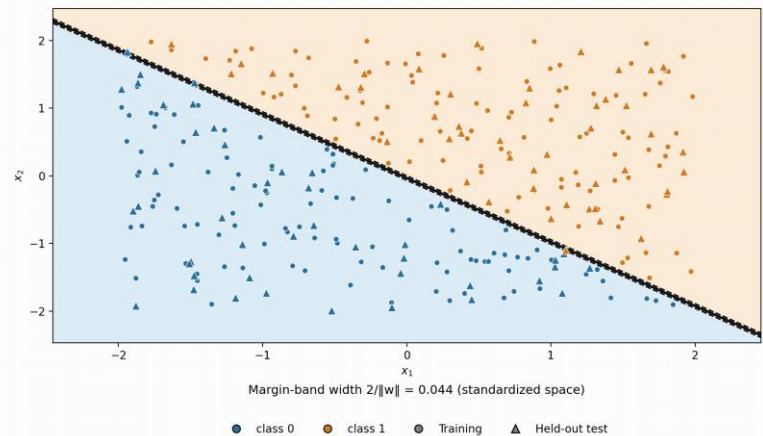
$$\begin{aligned} \min_{w,b,\xi} \quad & \frac{1}{2} \|w\|^2 + c \sum_{i=1}^n \xi^i \\ \text{s.t.} \quad & 1 - \xi^i - y^i(w^\top x^i + b) \leq 0 \quad \forall i, \\ & -\xi^i \leq 0 \quad \forall i. \end{aligned}$$



Support Vector Machine

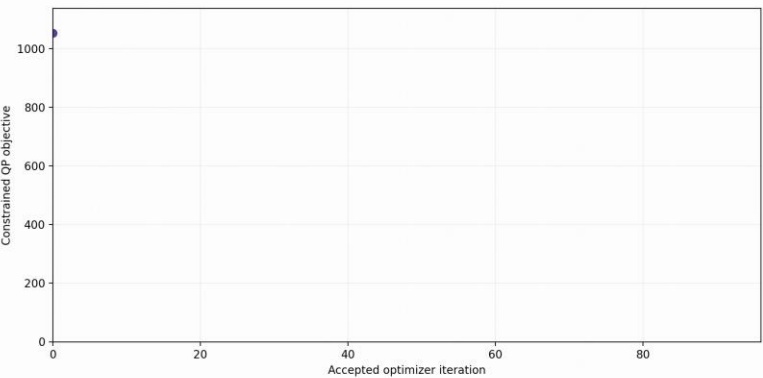
Synthetic: diagonal separation

Hard-margin SVM | Optimizer iteration 0 / 96
Solid: $f = 0$; dashed: $f = \pm 1$; colors: predicted class



Synthetic: diagonal separation

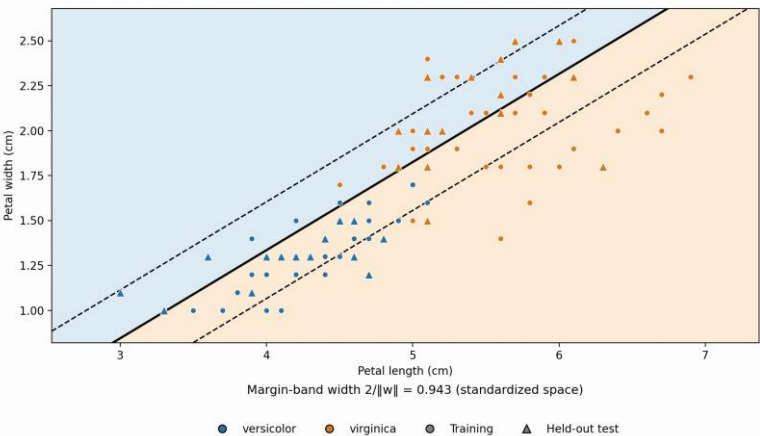
Hard-margin SVM | Optimizer iteration 0 / 96
Objective = 1054.0104



Feasible log-barrier Newton iterations; artificial barrier excluded from plotted objective.

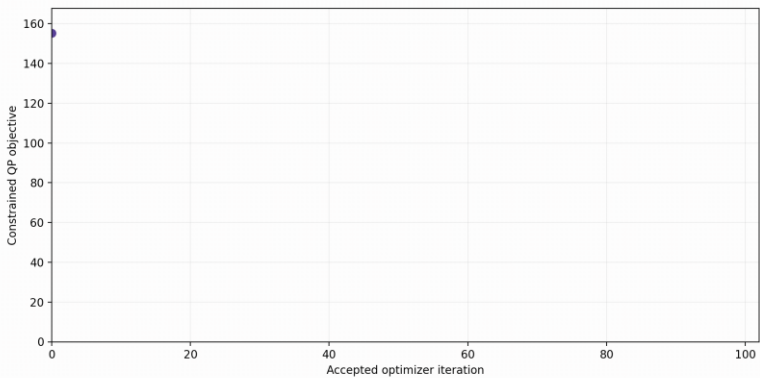
Iris: two species, two measurements

Soft-margin SVM ($C = 1$) | Optimizer iteration 0 / 102
Solid: $f = 0$; dashed: $f = \pm 1$; colors: predicted class



Iris: two species, two measurements

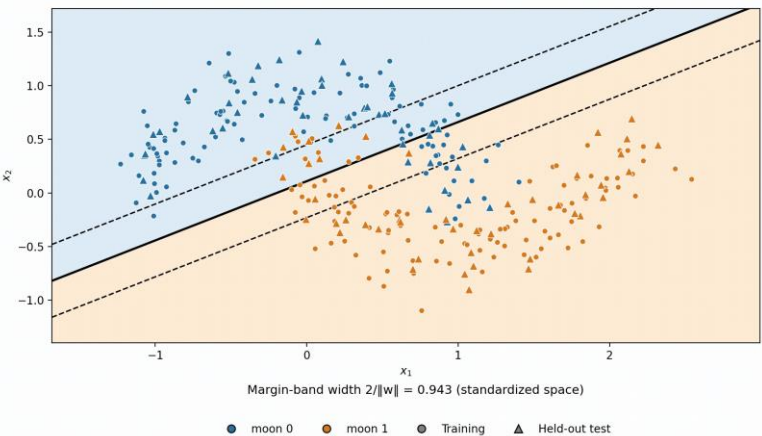
Soft-margin SVM ($C = 1$) | Optimizer iteration 0 / 102
Objective = 155.3358



Feasible log-barrier Newton iterations; artificial barrier excluded from plotted objective.

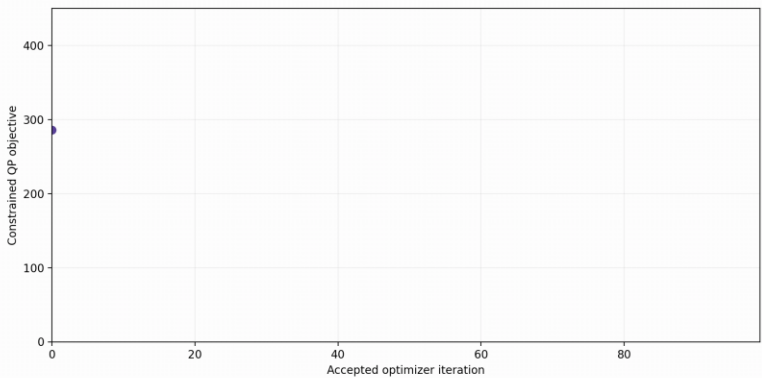
Synthetic: two interleaving moons

Soft-margin SVM ($C = 1$) | Optimizer iteration 0 / 99
Solid: $f = 0$; dashed: $f = \pm 1$; colors: predicted class



Synthetic: two interleaving moons

Soft-margin SVM ($C = 1$) | Optimizer iteration 0 / 99
Objective = 285.8587

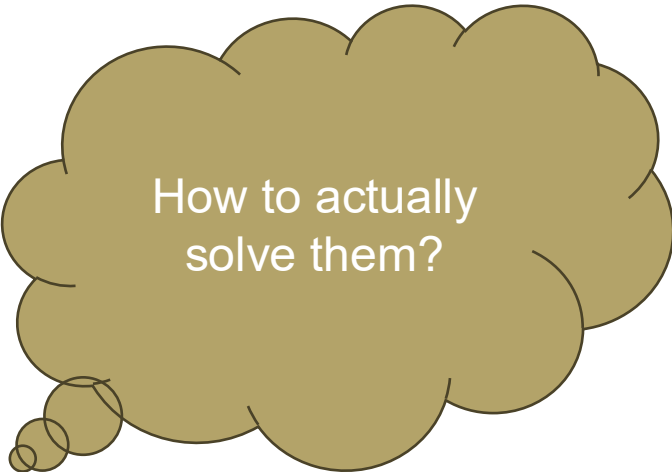


Feasible log-barrier Newton iterations; artificial barrier excluded from plotted objective.

Support Vector Machine

- Find decision boundary w as far from data point as possible

$$\min_{w,b} \frac{1}{2} \|w\|^2$$
$$s.t. y^i (w^\top x^i + b) \geq 1, \quad \forall i$$



How to actually solve them?

- Find decision boundary w as far from data point as possible (with some exceptions ξ)

$$\min_{w,b,\xi} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi^i$$
$$s.t. y^i (w^\top x^i + b) \geq 1 - \xi^i, \quad \xi^i \geq 0 \quad \forall i$$

Lagrangian Duality

Optimization and Lagrangian Duality

- The **primal** problem

$$\begin{aligned} & \min_w f(w) \\ \text{s.t. } & g_i(w) \leq 0, \quad i = 1, 2, \dots, k \\ & h_i(w) = 0, \quad i = 1, 2, \dots, l \end{aligned}$$

- The Lagrangian function

$$L(w, \alpha, \beta) = f(w) + \sum_{i=1}^k \alpha_i g_i(w) + \sum_{i=1}^l \beta_i h_i(w)$$

$\alpha_i \geq 0$ and β_i are called the Lagrangian multipliers

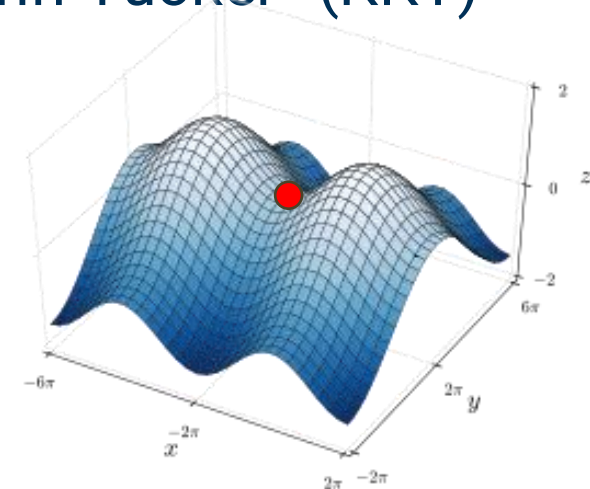
The Lagrangian and the KKT Conditions

- The Lagrangian function

$$L(w, \alpha, \beta) = f(w) + \sum_{i=1}^k \alpha_i g_i(w) + \sum_{i=1}^l \beta_i h_i(w)$$

- If w is an optimal solution to the primal problem, then there exist multipliers α, β (under some assumptions) such that the following “Karush-Kuhn-Tucker” (KKT) conditions hold:

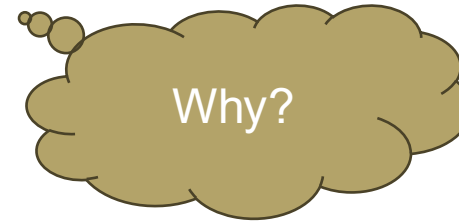
- | | |
|-----------------------------|-------------------------------------|
| 1. Stationarity: | $\frac{\partial L}{\partial w} = 0$ |
| 2. Primal feasibility: | $g_i(w) \leq 0, \quad h_i(w) = 0$ |
| 3. Dual feasibility: | $\alpha_i \geq 0$ |
| 4. Complementary slackness: | $\alpha_i g_i(w) = 0$ |



Properties of Lagrangian

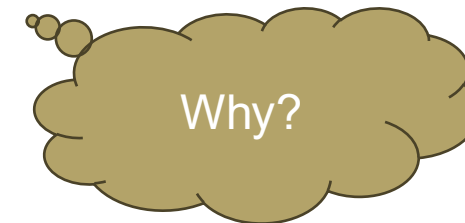
- For any feasible dual variables (Lagrangian multipliers) $\alpha_i \geq 0$ and β_i
- We have:

$$\min_{\substack{w \text{ feasible:} \\ g_i(w) \leq 0, h_i(w) = 0 \forall i}} f(w) \geq \inf_w L(w, \alpha, \beta)$$



- This is great! Can we find the best lower bound by optimizing over $\alpha_i \geq 0$ and β_i ?

$$\min_{\substack{w \text{ feasible:} \\ g_i(w) \leq 0, h_i(w) = 0 \forall i}} f(w) \geq \max_{\substack{\alpha, \beta \text{ feasible} \\ \alpha_i \geq 0 \forall i}} \inf_w L(w, \alpha, \beta)$$



Lagrangian Dual Problem

- **Dual problem:** maximizing the lower bound

$$\max_{\substack{\alpha, \beta \text{ feasible} \\ \alpha_i \geq 0 \forall i}} \inf_w L(w, \alpha, \beta)$$

- Dual objective: $g(\alpha, \beta) := \inf_w L(w, \alpha, \beta) = \inf_w f(w) + \sum_{i=1}^k \alpha_i g_i(w) + \sum_{i=1}^l \beta_i h_i(w)$
 - Dual variables: α, β
 - Dual constraints: $\alpha \geq 0$
- **Dual problem (in a different form):**

$$\begin{aligned} &\max_{\alpha, \beta} g(\alpha, \beta) \\ &s.t. \alpha \geq 0 \end{aligned}$$

Strong and Weak Duality

- **Primal problem:**

$$\begin{aligned} p^* &= \min_w f(w) \\ \text{s.t. } g_i(w) &\leq 0, & i = 1, 2, \dots, k \\ h_i(w) &= 0, & i = 1, 2, \dots, l \end{aligned}$$

- **Dual problem:** maximizing the lower bound

$$\begin{aligned} d^* &= \max_{\alpha, \beta \text{ feasible}} g(\alpha, \beta) \\ \text{s.t. } \alpha &\geq 0 \end{aligned}$$

- **Duality gap:** $p^* - d^* \geq 0$
 - **Weak duality:** we know that $p^* \geq d^*$ always holds
 - **Strong duality:** $p^* - d^* = 0$

Hard-Margin SVM Dual Problem

Hard-margin SVM

$$\min_{w,b} \frac{1}{2} \|w\|^2$$

$$s.t. 1 - y^i(w^\top x^i + b) \leq 0, \quad \forall i$$

- The Lagrangian function

$$L(w, b, \alpha) = \frac{1}{2} w^\top w + \sum_{i=1}^n \alpha_i (1 - y^i(w^\top x^i + b))$$

Deriving the Dual Problem

- $L(w, b, \alpha) = \frac{1}{2} w^\top w + \sum_{i=1}^n \alpha_i (1 - y^i (w^\top x^i + b))$
- **Dual objective:** $g(\alpha) := \inf_{w, b} L(w, b, \alpha)$
- Taking derivative and set to zero to find optimal w and b

$$\frac{\partial L}{\partial w} = w^* - \sum_{i=1}^n \alpha_i y^i x^i = 0$$
$$\frac{\partial L}{\partial b} = - \sum_{i=1}^n \alpha_i y^i = 0$$

Plug Back the Relation of w and b

$$\begin{aligned} g(\alpha) &:= L(w^*, b^*, \alpha) = \frac{1}{2} w^{*\top} w^* + \sum_{i=1}^n \alpha_i \left(1 - y^i (w^{*\top} x^i + b) \right) \\ &= \frac{1}{2} \left(\sum_{i=1}^n \alpha_i y^i x^i \right)^\top \left(\sum_{j=1}^n \alpha_j y^j x^j \right) + \sum_{i=1}^n \alpha_i \left(1 - y^i \left(\left(\sum_{j=1}^n \alpha_j y^j x^j \right)^\top x^i + b \right) \right) \\ &= \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^n \alpha_i \alpha_j y^i y^j (x^{i\top} x^j) - b \underbrace{\sum_{i=1}^n \alpha_i y^i}_0 \end{aligned}$$

The Dual Problem of SVM

$$\begin{aligned} \max_{\alpha} g(\alpha) &:= \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^n \alpha_i \alpha_j y^i y^j (x^i{}^\top x^j) \\ \text{s.t. } \alpha_i &\geq 0 \quad \forall i = 1, 2, \dots, m \\ \underbrace{\sum_{i=1}^n \alpha_i y^i}_{\text{Remember to put the constraints back!}} &= 0 \end{aligned}$$

- This is a constrained quadratic program
- Nice and **concave**, and global maximum can be found
- How to use the dual solution α to make prediction/classification?

Why convex?

Soft-Margin SVM Dual Problem

Soft-margin SVM

$$\begin{aligned} \min_{w,b,\xi} \quad & \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi^i \\ \text{s.t.} \quad & 1 - \xi^i - y^i(w^\top x^i + b) \leq 0 \quad \forall i, \\ & -\xi^i \leq 0 \quad \forall i. \end{aligned}$$

Deriving the Dual Problem

- Define Lagrangian:

$$L(w, b, \xi, \alpha, \beta) = \frac{1}{2}w^\top w + \sum_{i=1}^n C\xi^i + \alpha_i \left(1 - \xi^i - y^i(w^\top x^i + b)\right) - \beta_i \xi^i$$

- Taking derivative and set to zero:

$$\frac{\partial L}{\partial w} = w^* - \sum_{i=1}^n \alpha_i y^i x^i = 0$$

$$\frac{\partial L}{\partial b} = - \sum_{i=1}^n \alpha_i y^i = 0$$

$$\frac{\partial L}{\partial \xi^i} = C - \alpha_i - \beta_i = 0$$

Plug Back Relation of w , b , and ξ

$$\begin{aligned} g(\alpha, \beta) &:= \inf_{w, b, \xi} L(w, b, \xi, \alpha, \beta) = L(w^*, b^*, \xi^*, \alpha, \beta) \\ &= \frac{1}{2} \left(\sum_{i=1}^n \alpha_i y^i x^i \right)^\top \left(\sum_{i=1}^n \alpha_i y^i x^i \right) + \sum_{i=1}^n C \xi^i + \alpha_i \left(1 - \xi^i - y^i \left(\left(\sum_{i=1}^n \alpha_i y^i x^i \right)^\top x^i + b \right) \right) - \beta_i \xi^i \\ &= \frac{1}{2} \left(\sum_{i=1}^n \alpha_i y^i x^i \right)^\top \left(\sum_{j=1}^n \alpha_j y^j x^j \right) + \sum_{i=1}^n \alpha_i \left(1 - y^i \left(\left(\sum_{j=1}^n \alpha_j y^j x^j \right)^\top x^i + b \right) \right) \end{aligned}$$

- After simplification

$$g(\alpha, \beta) := \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^n \alpha_i \alpha_j y^i y^j (x^i{}^\top x^j)$$

The Dual Problem

$$\begin{aligned} \max_{\alpha, \beta} g(\alpha, \beta) &:= \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^n \alpha_i \alpha_j y^i y^j (x^i{}^\top x^j) \\ \text{s.t. } C - \alpha_i - \beta_i &= 0, \quad \alpha_i \geq 0, \quad \beta_i \geq 0 \quad \forall i \\ \sum_{i=1}^n \alpha_i y^i &= 0 \end{aligned}$$

- The constraint $C - \alpha_i - \beta_i = 0$, $\alpha_i \geq 0$, $\beta_i \geq 0$ can be simplified to $C \geq \alpha_i \geq 0$
- This is a constrained quadratic programming
- Nice and concave, and global maximum can be found

The Dual Problem

$$\begin{aligned} \max_{\alpha} g(\alpha) &:= \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^n \alpha_i \alpha_j y^i y^j (x^i{}^\top x^j) \\ \text{s. t. } &\frac{C}{n} \geq \alpha_i \geq 0 \quad \forall i \\ &\sum_{i=1}^n \alpha_i y^i = 0 \end{aligned}$$

- **Intuition:**

- When $C \rightarrow \infty$, then this becomes the hard margin SVM.
- Nice and concave, and global maximum can be found

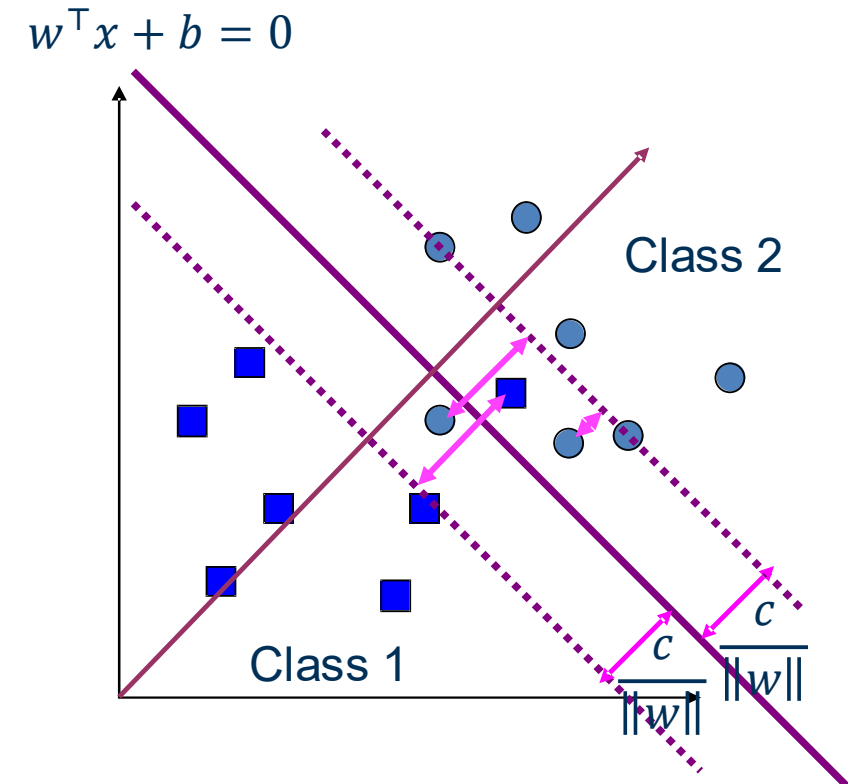
Hard-margin SVM

$$\begin{aligned} \max_{\alpha} g(\alpha) &:= \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^n \alpha_i \alpha_j y^i y^j (x^i{}^\top x^j) \\ \text{s. t. } &\alpha_i \geq 0 \quad \forall i = 1, 2, \dots, m \\ &\sum_{i=1}^n \alpha_i y^i = 0 \end{aligned}$$

Inference and Support Vectors

Complementary Slackness

- What is the geometric meaning of α_i ?
- Complementary slackness
 - $\alpha_i (1 - \xi^i - y^i(w^\top x^i + b)) = 0$
 - $(C - \alpha_i) \xi_i = 0$. (because $\beta_i = C - \alpha_i$)



Primal problem

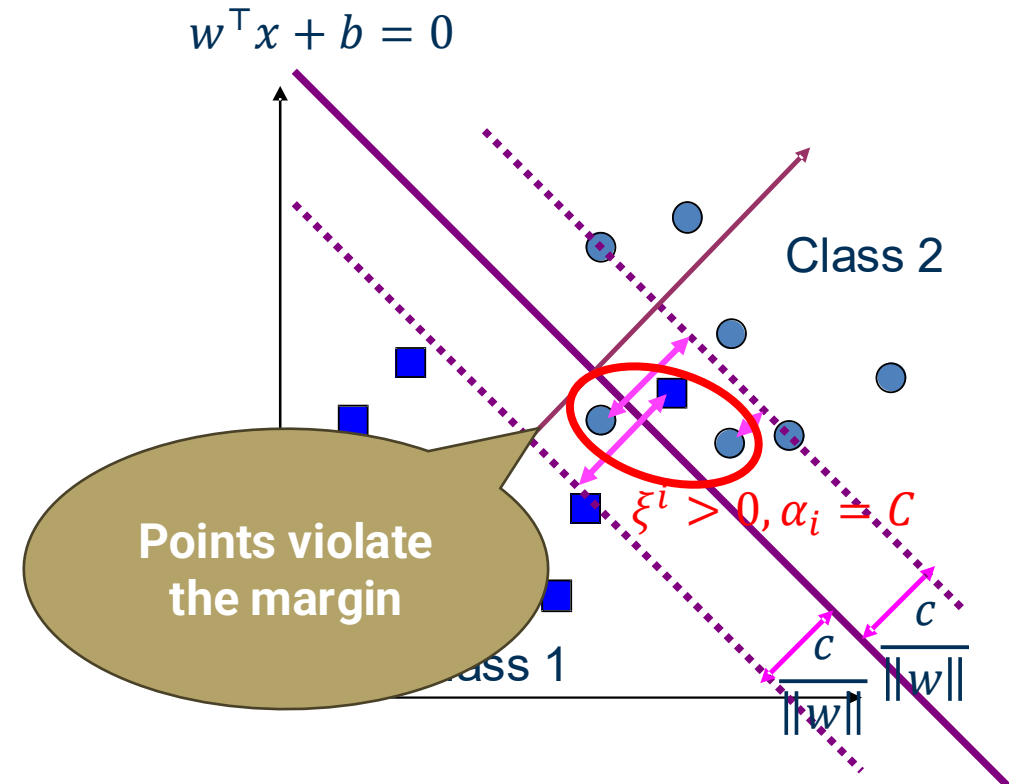
$$\begin{aligned} \min_{w, b, \xi} \quad & \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi^i \\ \text{s.t.} \quad & 1 - \xi^i - y^i(w^\top x^i + b) \leq 0, \\ & -\xi^i \leq 0 \quad \forall i \end{aligned}$$

Dual problem

$$\begin{aligned} \max_{\alpha} \quad & g(\alpha) := \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^n \alpha_i \alpha_j y^i y^j (x^{i\top} x^j) \\ \text{s.t.} \quad & C \geq \alpha_i \geq 0 \quad \forall i \\ & \sum_{i=1}^n \alpha_i y^i = 0 \end{aligned}$$

Complementary Slackness

- **Non-margin support vectors $\alpha_i = C \neq 0$:**
 - This means $\beta_i = C - \alpha_i = 0$
 - Therefore, $\xi_i \geq 0$, and $1 - y^i(w^\top x^i + b) = \xi_i \geq 0$



Primal problem

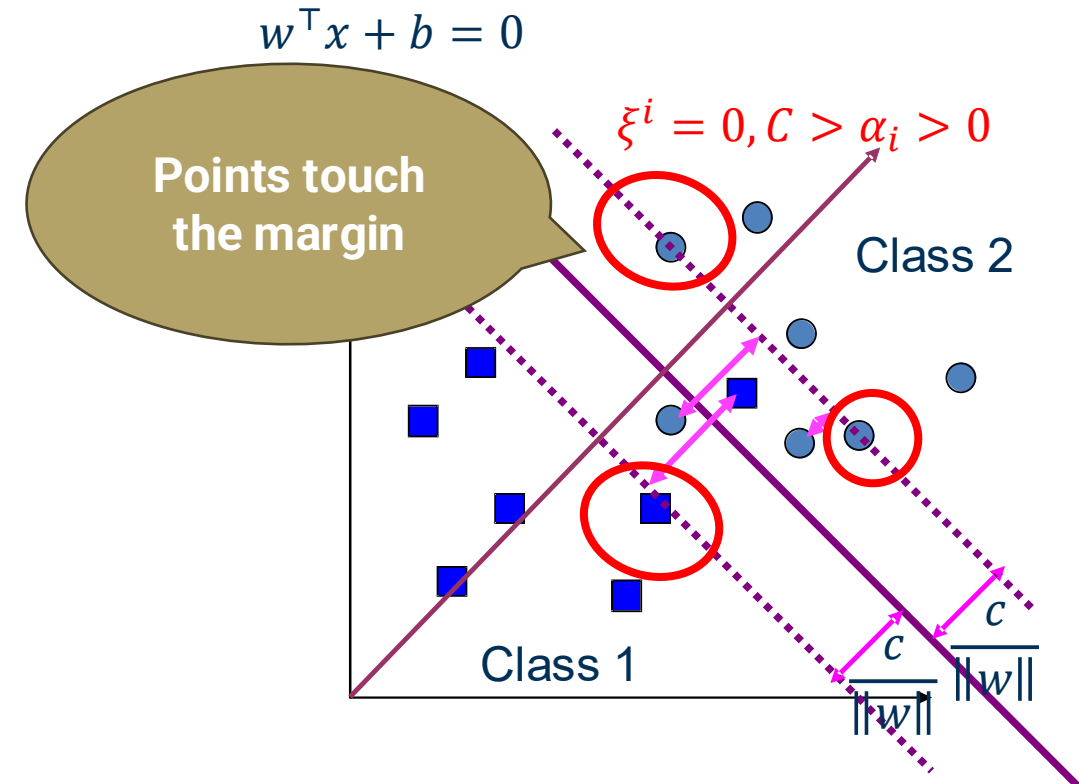
$$\begin{aligned} \min_{w, b, \xi} \quad & \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i \\ \text{s.t.} \quad & 1 - \xi_i - y^i(w^\top x^i + b) \leq 0, \\ & -\xi_i \leq 0 \quad \forall i \end{aligned}$$

Dual problem

$$\begin{aligned} \max_{\alpha} \quad & g(\alpha) := \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^n \alpha_i \alpha_j y^i y^j (x^{i\top} x^j) \\ \text{s.t.} \quad & C \geq \alpha_i \geq 0 \quad \forall i \\ & \sum_{i=1}^n \alpha_i y^i = 0 \end{aligned}$$

Complementary Slackness

- **Non-margin support vectors $\alpha_i = C \neq 0$:**
 - This means $\beta_i = C - \alpha_i = 0$
 - Therefore, $\xi_i \geq 0$, and $1 - y^i(w^\top x^i + b) = \xi_i \geq 0$
- **Margin support vectors $0 < \alpha_i < C$:**
 - This means $\beta_i = C - \alpha_i > 0$
 - Therefore, $\xi_i = 0$, and $1 - y^i(w^\top x^i + b) = \xi_i = 0$



Primal problem

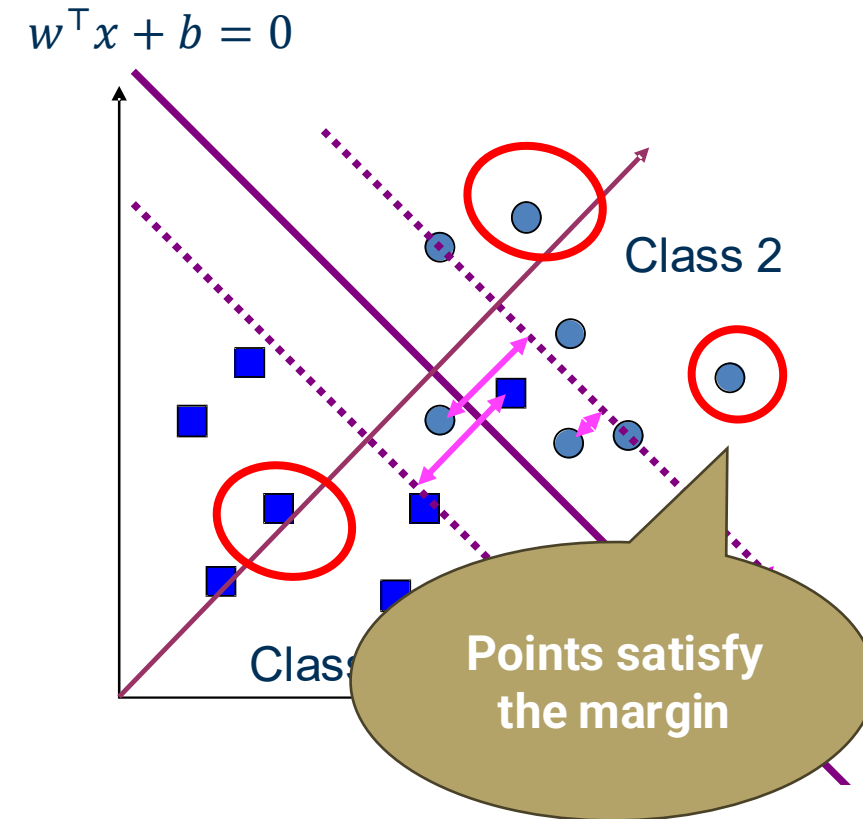
$$\begin{aligned} \min_{w, b, \xi} \quad & \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i \\ \text{s.t.} \quad & 1 - \xi_i - y^i(w^\top x^i + b) \leq 0, \\ & -\xi_i \leq 0 \quad \forall i \end{aligned}$$

Dual problem

$$\begin{aligned} \max_{\alpha} \quad & g(\alpha) := \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^n \alpha_i \alpha_j y^i y^j (x^i{}^\top x^j) \\ \text{s.t.} \quad & C \geq \alpha_i \geq 0 \quad \forall i \\ & \sum_{i=1}^n \alpha_i y^i = 0 \end{aligned}$$

Complementary Slackness

- **Non-margin support vectors $\alpha_i = C \neq 0$:**
 - This means $\beta_i = C - \alpha_i = 0$
 - Therefore, $\xi_i \geq 0$, and $1 - y^i(w^\top x^i + b) = \xi_i \geq 0$
- **Margin support vectors $0 < \alpha_i < C$:**
 - This means $\beta_i = C - \alpha_i > 0$
 - Therefore, $\xi_i = 0$, and $1 - y^i(w^\top x^i + b) = \xi_i = 0$
- **Non-support vectors $0 = \alpha_i$:**
 - This means $\beta_i = C - \alpha_i = C > 0$
 - Therefore, $\xi_i = 0$, i.e., $1 - y^i(w^\top x^i + b) \leq \xi_i = 0$



Classification by Support Vectors

- Recall that we have:

$$\mathbf{w} = \sum_{i=1}^n \alpha_i y^i x^i$$

- Pick any data point with $0 < \alpha_i < C$ to compute b by:

$$1 - y^i (\mathbf{w}^\top x^i + b) = 0$$

- For a new test point z

- Compute

$$\mathbf{w}^\top z + b = \left(\sum_{i=1}^n \alpha_i y^i (x^i{}^\top z) \right) + b = \left(\sum_{i \in \text{support vectors}} \alpha_i y^i (x^i{}^\top z) \right) + b$$

- Classify z as class 0 if the result is negative (or $\mathbf{w}^\top x + b \leq 0$)
- Classify z as class 1 if the result is positive (or $\mathbf{w}^\top x + b \geq 0$)

Inference of Support Vector Machines

- We make decisions by comparing each new example z with only the support vectors:

$$y^* = \text{sign}(w^\top z + b) = \text{sign} \left(\left(\sum_{i \in \text{support vectors}} \alpha_i y^i (x^{i^\top} z) \right) + b \right)$$

Primal

- Primal solution w^*, b^*
- Inference: $\text{sign}(w^\top z + b)$
- Computation cost: $O(d)$

Dual

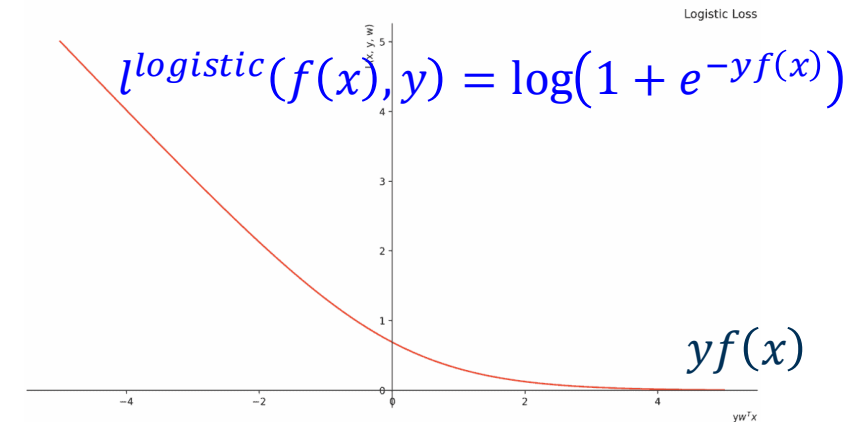
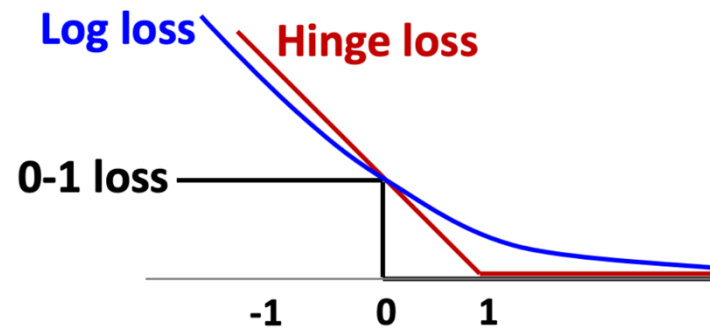
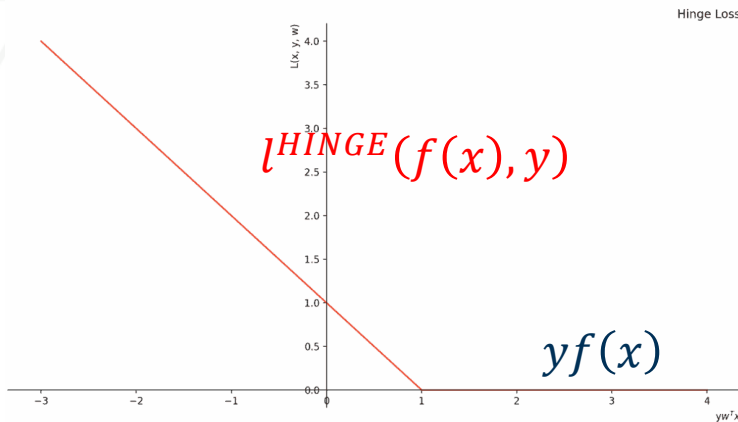
- Dual solution α with n' support vectors (nonzero α_i)
- Inference: $\text{sign} \left(\left(\sum_{i \in \text{support vectors}} \alpha_i y^i (x^{i^\top} z) \right) + b \right)$
 - Need to pre-compute the offset $b \in \mathbb{R}$ by any support vector
- Computation cost: $O(dn')$

Hinge Loss

Hinge Loss

- **Definition:** Assuming the label $y \in \{-1, 1\}$ and the decision rule is $h(x) = \text{sign}(f(x))$ with $f(x) = w^\top x + b$

$$l^{\text{HINGE}}(f(x), y) = \begin{cases} 0 & \text{if } yf(x) \geq 1 \\ 1 - yf(x) & \text{otherwise} \end{cases}$$



- **Intuition:** penalize more if incorrectly classified (the left side of the curve)
- **Convenient shorthand:**

$$l^{\text{HINGE}}(f(x), y) = \max(0, 1 - yf(x))$$

Equivalence of Hinge Loss to SVMs

- Minimizing the total hinge loss on all the training data

$$\min_{w,b} \sum_{i=1}^n \max(0, 1 - y^i [w^\top x^i + b]) + \frac{\lambda}{2} \|w\|^2$$

Hinge loss

Regularization

- This is analogous to the regularized least square, which balances two terms (the loss and the regularizer). Conventionally, we rewrite the objective function as ($C = \frac{1}{\lambda}$):

$$\min_{w,b} C \sum_{i=1}^n \max(0, 1 - y^i [w^\top x^i + b]) + \frac{1}{2} \|w\|^2$$

- We further rewrite this into another equivalent form

$$\begin{aligned} \min_{w,b,\xi} C \sum_{i=1}^n \xi^i + \frac{1}{2} \|w\|^2 \\ \text{s.t. } \max(0, 1 - y^i [w^\top x^i + b]) = \xi^i \quad \forall i \end{aligned}$$

Equivalence of Hinge Loss to SVMs

- **Primal formulation:**

$$\begin{aligned} \min_{w,b,\xi} \quad & C \sum_{i=1}^n \xi^i + \frac{1}{2} \|w\|^2 \\ \text{s.t.} \quad & 1 - y^i [w^\top x^i + b] \leq \xi^i \quad \forall i \\ & 0 \leq \xi^i \end{aligned}$$

Where all ξ^i are called slack variables.

Soft-margin SVM

$$\begin{aligned} \min_{w,b,\xi} \quad & \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi^i \\ \text{s.t.} \quad & y^i (w^\top x^i + b) \geq 1 - \xi^i, \\ & \xi^i \geq 0 \quad \forall i \end{aligned}$$

Kernel (nonlinear) SVM

The Dual Problem

$$\begin{aligned} \max_{\alpha} g(\alpha) &:= \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^n \alpha_i \alpha_j y^i y^j (x^i{}^\top x^j) \\ \text{s. t. } C &\geq \alpha_i \geq 0 \quad \forall i \\ \sum_{i=1}^n \alpha_i y^i &= 0 \end{aligned}$$

Hard-margin SVM

$$\begin{aligned} \max_{\alpha} g(\alpha) &:= \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^n \alpha_i \alpha_j y^i y^j (x^i{}^\top x^j) \\ \text{s. t. } \alpha_i &\geq 0 \quad \forall i = 1, 2, \dots, m \\ \sum_{i=1}^n \alpha_i y^i &= 0 \end{aligned}$$

Kernel SVM

$$\begin{aligned} \max_{\alpha} g(\alpha) &:= \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^n \alpha_i \alpha_j y^i y^j K(x^i, x^j) \\ \text{s. t. } &\textcolor{red}{C} \geq \alpha_i \geq 0 \quad \forall i \\ &\sum_{i=1}^n \alpha_i y^i = 0 \end{aligned}$$

Hard-margin kernel SVM

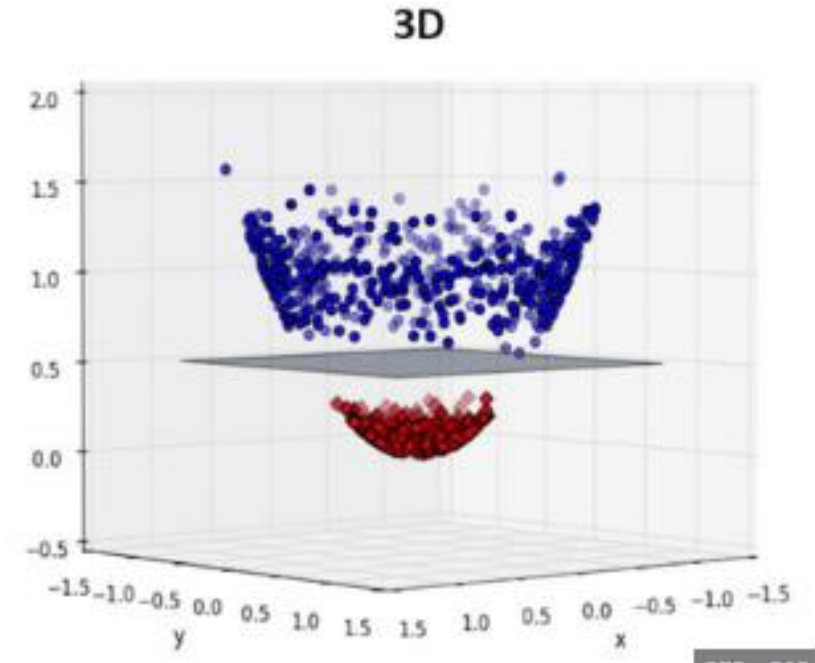
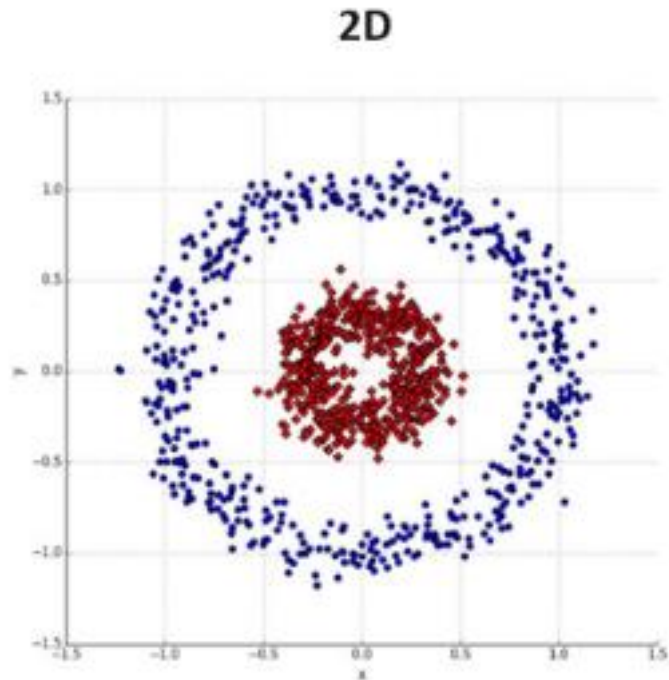
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Kernel SVM

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- Kernel $K(x, x')$ measures the similarity of any pair $x, x' \in X$
 - **General kernel:** $K: X \times X \rightarrow \mathbb{R}$
 - **Inner product-like kernel:** $K(x, x') = \langle \phi(x), \phi(x') \rangle_V$, where $\phi: X \rightarrow V$
- Map the data points x^i into a higher-dimensional space $\phi(x^i)$ where the separation becomes easier.

Kernel SVM

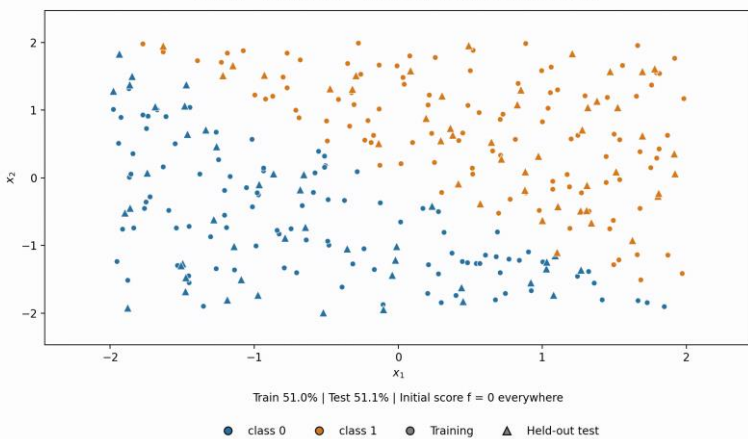


- What kernel to use in SVM?
 - Kernel $K(x, x')$ measures the similarity between two points $x, x' \in X$
 - Pick your similarity measure!

Kernel SVM Demo

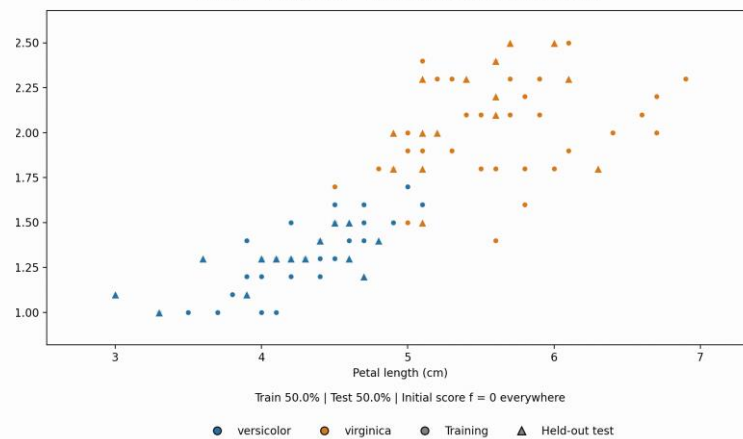
Synthetic: diagonal separation

RBFSVM ($C = 1$, $\gamma = 1$) | SMO iteration 0 / 400
Solid: $f = 0$; dashed: $f = \pm 1$; rings: nonzero dual coefficients



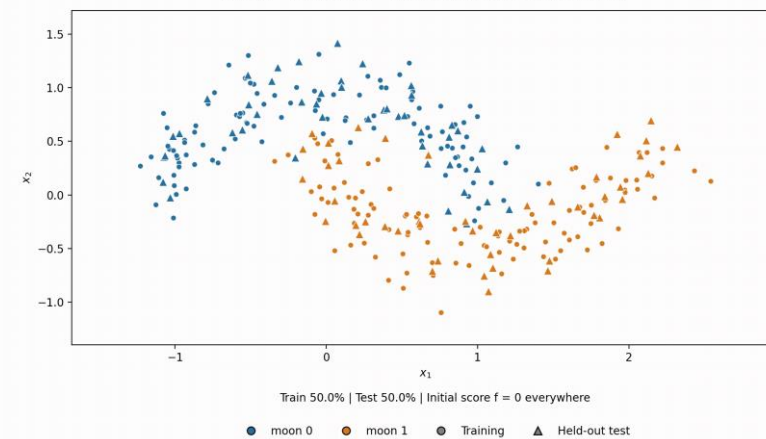
Iris: two species, two measurements

RBFSVM ($C = 1$, $\gamma = 1$) | SMO iteration 0 / 93
Solid: $f = 0$; dashed: $f = \pm 1$; rings: nonzero dual coefficients



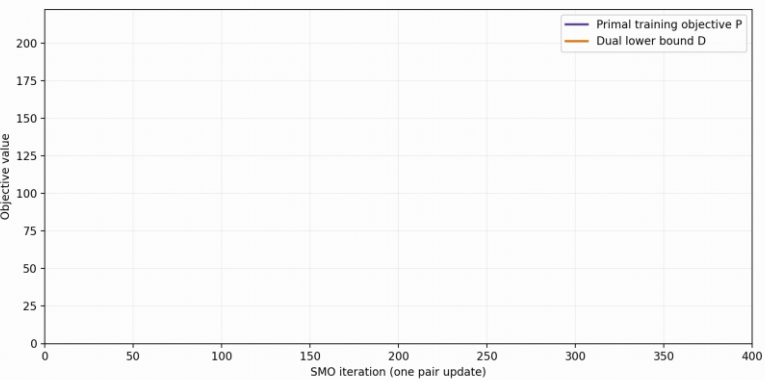
Synthetic: two interleaving moons

RBFSVM ($C = 1$, $\gamma = 1$) | SMO iteration 0 / 234
Solid: $f = 0$; dashed: $f = \pm 1$; rings: nonzero dual coefficients



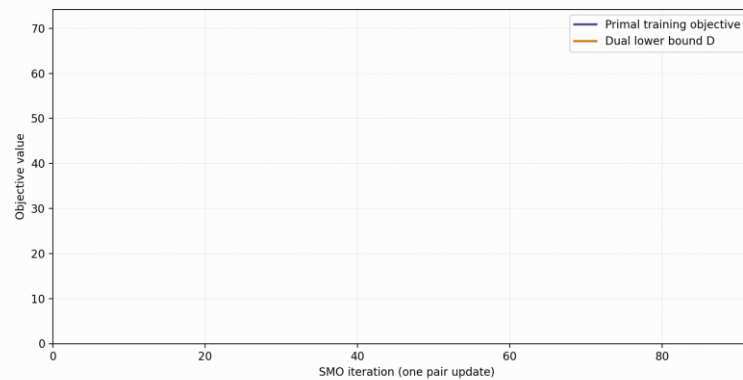
Synthetic: diagonal separation

RBFSVM | SMO iteration 0 / 400
Primal = 210.0000 | Dual = 0.0000 | Gap = 210



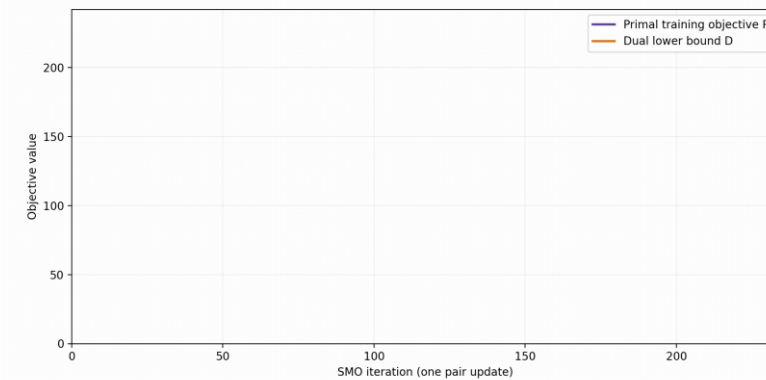
Iris: two species, two measurements

RBFSVM | SMO iteration 0 / 93
Primal = 70.0000 | Dual = 0.0000 | Gap = 70



Synthetic: two interleaving moons

RBFSVM | SMO iteration 0 / 234
Primal = 210.0000 | Dual = 0.0000 | Gap = 210



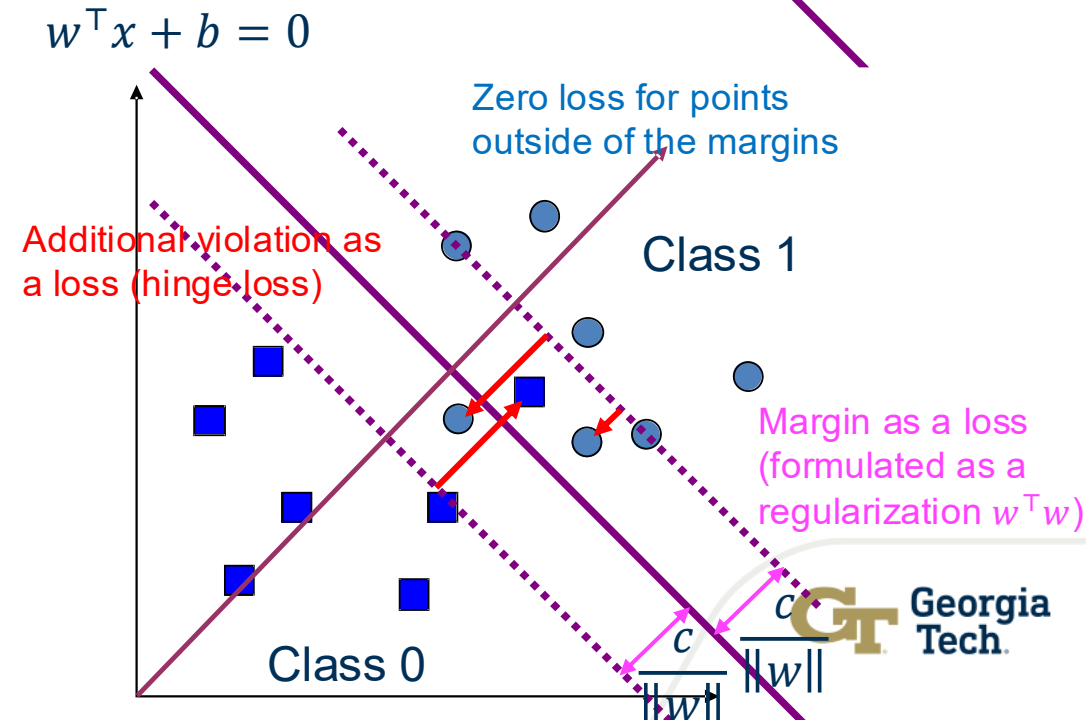
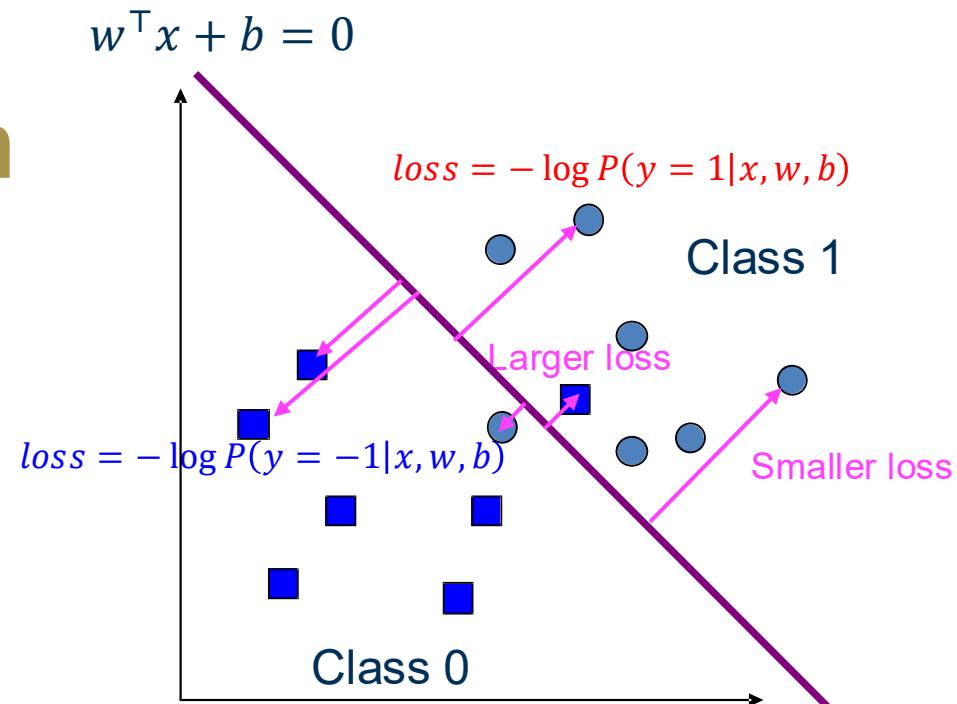
Advantage of Dual SVM

- Short answer: **kernel**
- Long answer: **kernel**
- **Primal**: no good kernel extension
- **Dual**: naturally extend to different kernels
- **Primal**: learns w, b and classifies z by computing $w^\top z + b$
- **Dual**: learns α and classifies z by computing $\sum_{i \in \text{support vectors}} \alpha_i y^i (x^i{}^\top z) + b$

Comparisons and Takeaways

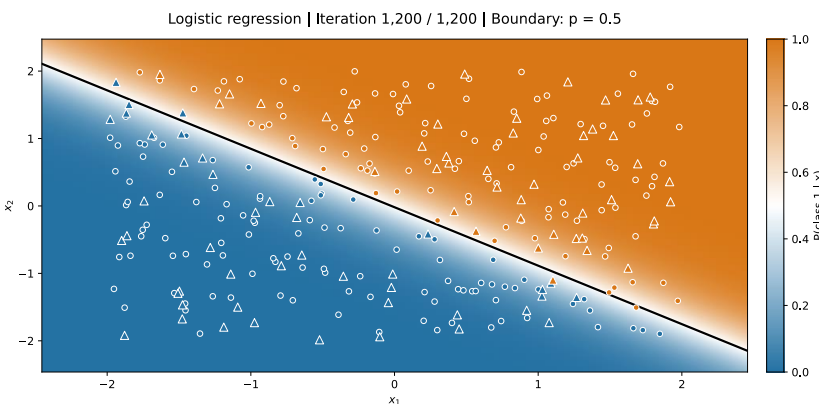
SVMs v.s. Logistic Regression

- **Logistic regression** focuses on maximizing the probability of the data. The farther the data lies from the separating hyperplane (on the correct side), the happier LR is.
- **SVM** tries to find the separating hyperplane that maximizes the distance of the closest points to the margin (the support vectors). If a point is not a support vector, it doesn't really matter.

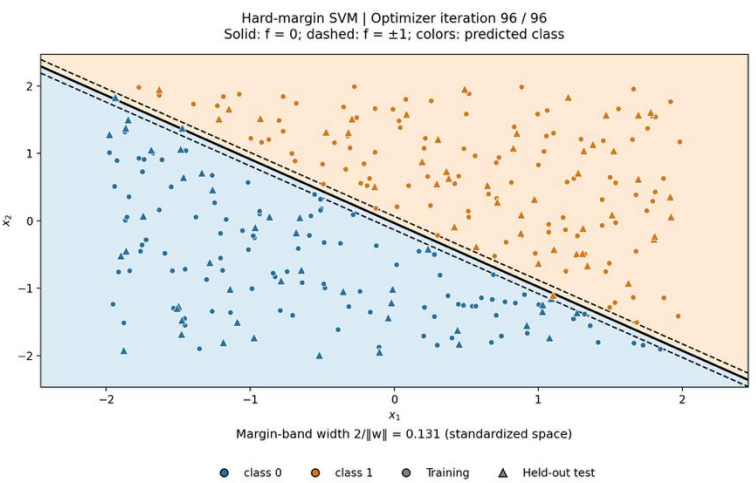


SVMs v.s. Logistic Regression

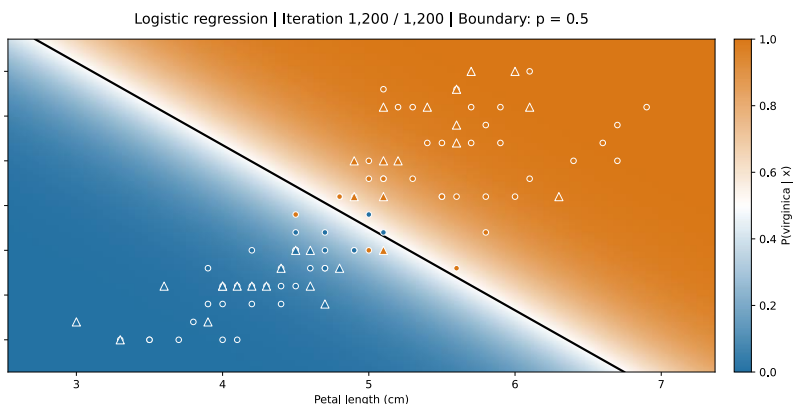
Synthetic: diagonal separation



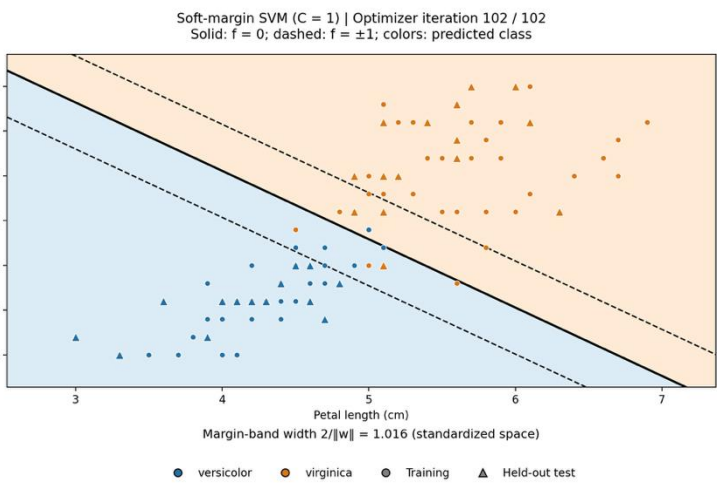
Synthetic: diagonal separation



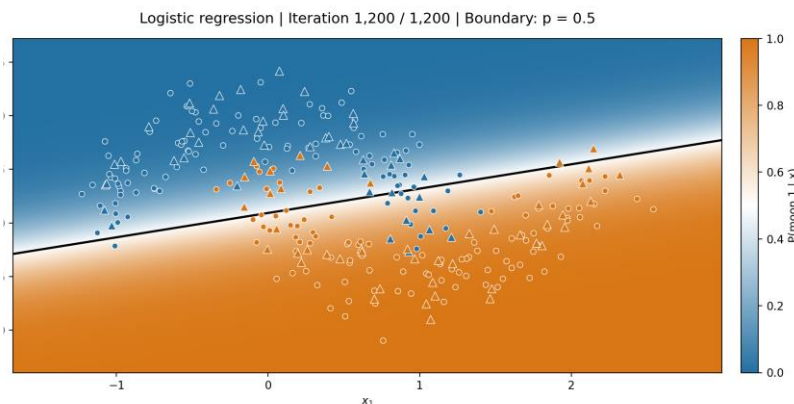
Iris: two species, two measurements



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Synthetic: two interleaving moons



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